EurekaZero: Supplementary Telemetry

Candidate movement precedes admission; retention and independent learning remain separate questions.

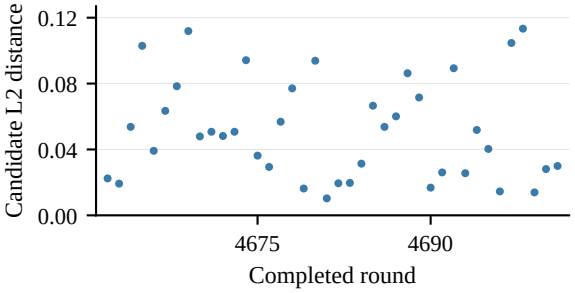


Figure S1: **Recorded candidate displacement varies across rounds.** All 40 complete retained summaries, rounds 4662–4701, captured at 00:55 UTC on 4 November 2026. Points show logged candidate parameter-space L2 displacement; no smoothing or filtering is applied. In the inspected loop, it includes warm-restart perturbation and optimization before admission. The journal records zero admissions in this window. This does not isolate optimizer effects or establish a retained learning gain. Scope: the adaptive adapter admission record in the main manuscript.

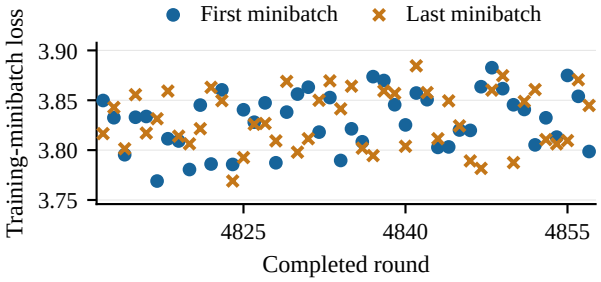


Figure S2: **First and last recorded minibatch losses use different batches.** All 46 complete retained summaries, rounds 4812–4857, captured at 05:53 UTC on 4 November 2026. The 92 points show logged first and last training-minibatch objectives, with two optimizer steps per candidate round. Batch-order hashes differ within every round; these are not matched before/after losses. No smoothing or filtering is applied. The window records zero admissions; these losses establish no retained learning gain. Scope: the adaptive adapter admission record in the main manuscript.

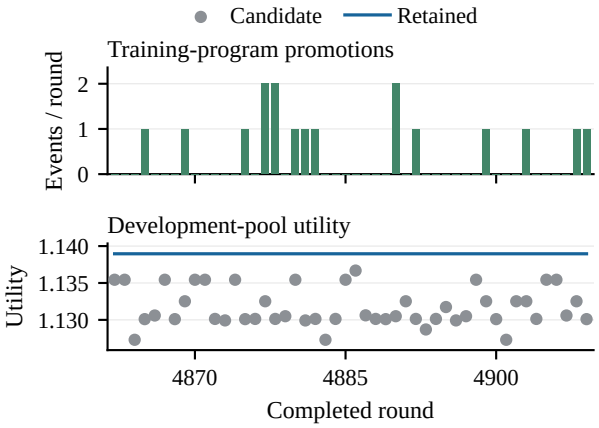


Figure S3: **Program promotion and adapter retention are distinct events.** All 48 complete retained summaries, rounds 4862–4909, captured at 07:21 UTC on 4 November 2026. Top: 17 strict training-program promotions, independently matched to the program journal; zero-event rounds retained. Bottom: candidate and post-decision retained adapter utility (higher is better); zero adapter admissions. Values are unsmoothed, without outcome filtering. Reused development contexts do not establish independent writer or ranking-map learning gains. The inspected caller decides adapter admission before program search; round alignment does not identify a training effect. Scope: the adaptive adapter admission record in the main manuscript.

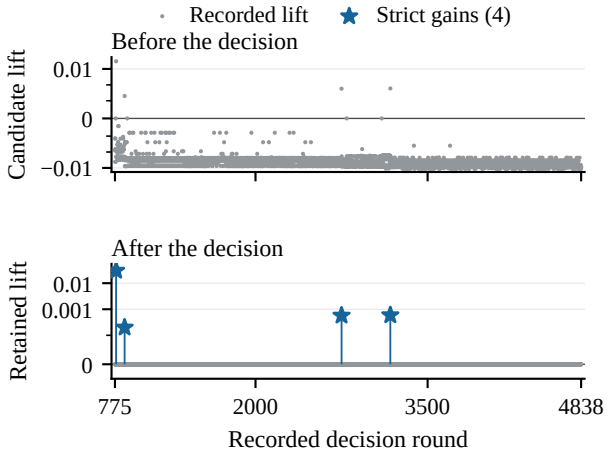


Figure S4: **Only four recorded decisions increase retained utility.** All 4,063 admission records in rounds 775–4838, captured at 04:56 UTC on 4 November 2026. Top: candidate utility minus the logged prior incumbent; bottom: post-decision retained utility minus that same incumbent. Stars mark the four strict gains. The other 4,059 retained differences are zero, including four accepted ties; zero does not imply unchanged weights. Axes are symmetric-log, linear within 10^{-4} utility units of zero. Points retain every observed decision; missing round 4331 is not imputed. These unsmoothed differences on reused development contexts establish neither a causal training effect nor independent writer or ranking-map learning. Scope: the adaptive adapter admission record in the main manuscript.

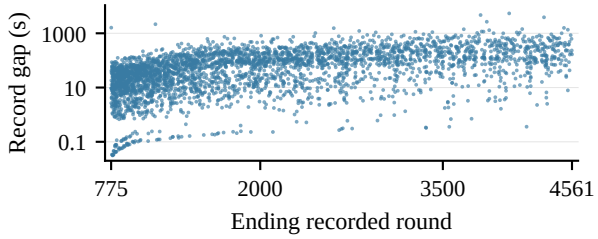


Figure S5: **Program-journal records arrive at varying intervals.** All 3,363 adjacent timestamp differences between 3,364 committed records in rounds 775–4561, captured at 20:57 UTC on 3 November 2026; warm-up records are excluded. Each point is keyed to the later record's round; all pairs are retained. The vertical axis is logarithmic. Recording cadence includes work, waits and batched writes, not per-proposal runtime or adapter admission. These observations establish neither a speed comparison nor learning. Scope: the training-program journal in the main manuscript.